

Mohamad Dikshie Fauzie

PRINCIPAL LOW-LATENCY INFRASTRUCTURE ARCHITECT · ELECTRONIC TRADING & DISTRIBUTED SYSTEMS

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Summary

Principal Low-Latency Infrastructure Architect and Systems Engineer with 10+ years of mission-critical production leadership co-managing physical, network, and storage infrastructure for Japan's largest Proprietary Trading System (Japannext PTS), operating under strict Japan FSA regulatory oversight. Holds a Ph.D. from Keio University in Distributed Systems (supervised by Internet Hall of Fame founder Prof. Jun Murai and quantum pioneer Prof. Rodney Van Meter) and a B.Sc. in Physics from ITB. Proven track record executing zero-downtime exchange datacenter migrations (delivered 2 weeks early, 50% expansion headroom), 50% rack power balancing optimization, and 50% I/O throughput gains on Dell PowerScale (Isilon) for multi-petabyte market tick data capture and replay. Specializes in sub-nanosecond precision timing (NPL UK Time & Frequency metrology, IEEE 1588v2 PTP, White Rabbit, SyncE), low-latency switching (Arista 7130), kernel-bypass networking (Solarflare, NetFPGA, DPDK), and high-density colocation across Tokyo exchanges (TY3, CC2). Certified in NYIF Financial Risk Management. Trilingual: English, Japanese, Indonesian.

Skills

Trading Systems	Exchange Colocation (TY3, CC2), Low-Latency Switching (Arista 7130 / MetaWatch), Kernel Bypass (Solarflare Onload, EF_VI, DPDK, AF_XDP), NetFPGA, Multicast (PIM-SM/SSM), Market Data (ITCH, FAST, BBO), Order Entry (OUCH, FIX)
Precision Timing	IEEE 1588v2 PTP (Grandmasters, Boundary/Transparent Clocks), SyncE, White Rabbit, Sub-Nanosecond Metrology, Atomic Clocks (Rubidium, OCXO), FSA / MiFID II RTS 25 Timestamp Compliance, Hardware Timestamping
Systems & OS	Linux Kernel Tuning (isolcpus, CPU Pinning, IRQ Affinity, HugePages, sysctl Network Stack Tuning), RHEL/CentOS, FreeBSD, High-Performance Compute (HPC) Clusters, Bare-Metal Provisioning
Storage & Cloud	Dell PowerScale (Isilon) Parallel Storage (High-Throughput Tick Data Capture & Backtesting), Kubernetes, Docker, OpenStack, NVIDIA AI Infrastructure (DLI Certified), AWS Cloud Fundamentals
Facilities & DC	Rack-Level PDU Power Telemetry, High-Density Thermal Balancing, Zero-Downtime Migration, Optical Physical Layer (DWDM, Single-Mode Fiber, OTDR, SFP/QSFP Transceiver Diagnostics), Colocation Cage Operations
Tools & Languages	Python, Bash/Shell, C/C++ familiarity, Git, SNMP, Prometheus/Grafana, LaTeX · English (Fluent), Japanese (Working), Indonesian (Native)

Work Experience

Japannext Co., Ltd. (Largest Proprietary Trading System / PTS in Japan)

Tokyo, Japan

INFRASTRUCTURE LEAD (CO-MANAGER) / LOW-LATENCY SYSTEMS ARCHITECT

Oct. 2014 - Present

- Co-managed 24/7 mission-critical, ultra-low-latency datacenter and network infrastructure for Japan's primary alternative trading venue, maintaining 99.999% availability under strict Japan Financial Services Agency (FSA) compliance.
- Directed end-to-end server room migration for power-constrained trading systems; executed zero-downtime cutover 2 weeks ahead of schedule, unlocking 50% expansion capacity for low-latency matching engines and participant gateways.
- Spearheaded rack-level PDU power telemetry across high-density trading server racks, optimizing workload power balancing by 50% and eliminating breaker-trip risks during peak market volatility sessions.
- Architected Dell PowerScale (Isilon) enterprise parallel storage, unlocking a 50% I/O throughput increase over legacy SAN/NAS to support multi-terabyte daily tick data logging, order-book replay, and quantitative backtesting.
- Engineered automated inventory and optical diagnostics platform for consumables (SFPs/QSFPs, patch panels, single-mode fiber), drastically accelerating fault isolation and MTTR on participant exchange cross-connects.
- Supervised datacenter room buildouts and low-latency cabling topologies for matching engines and broker colocation cages (Equinix TY3, ATTokyo CC2), reaching operational readiness ahead of schedule.
- Collaborated with Unix engineering to deploy real-time hardware telemetry and alerting for packet drops, optical link degradation, and thermal throttles across mission-critical execution nodes.
- Automated executive and regulatory capacity, power, and SLA compliance reporting using time-series pipelines, ensuring 100% audit readiness under Japan FSA regulatory requirements.

Keio University - JICA JST Indo Project

Tokyo, Japan

RESEARCH ASSISTANT (DISTRIBUTED SYSTEMS & NETWORKS)

Apr. 2011 - Sep. 2014

- Researched distributed network topologies, packet dissemination efficiency, and energy modeling across large-scale P2P systems and CDNs.
- Applied game theory and probabilistic gossip protocols to model message propagation, analyzing dynamic graph topologies under churn.
- Published peer-reviewed research papers in international IEEE and IEICE conferences and journals.

Keio University

Tokyo, Japan

SYSTEM ADMINISTRATOR (CAMPUS CORE INFRASTRUCTURE)

Oct. 2007 - Sep. 2014

- Administered campus server and core backbone network infrastructure (BGP/OSPF routing, DNS, email, switches, firewalls) with high uptime.

Certifications & Specialized Training

2022	Introduction to Time and Frequency Measurement (PTP / Clock Metrology) , National Physical Laboratory (NPL), UK	Teddington, UK
2020	Introduction to Metrology (Measurement Standards & Uncertainty) , National Physical Laboratory (NPL), UK	Teddington, UK
2010	Introduction to NetFPGA (High-Speed Line-Rate Packet Processing) , Stanford University	Tokyo, Japan
2022	Financial Risk Management & Capital Markets , New York Institute of Finance / Interactive Brokers	Online
2024	Foundations of Project Management , Google Career Certificates	Online
2024	AI Infrastructure and Operations Fundamentals , NVIDIA (Deep Learning Institute)	Online
2019	Cloud Infrastructure, Kubernetes & OpenStack , The Linux Foundation	Online
2023	AWS Certified Cloud Practitioner , Amazon Web Services (AWS)	Online
2019	Network Transformation: 5G, Core Network, Cloud & NFV , Intel	Online
2022 - 2023	Machine Learning & Mathematics for Machine Learning , Stanford University / DeepLearning.AI	Online

Publications

Energy Consumption in Peer Assisted CDN

*International Journal of Advanced
Research in Computer Science
(IJARCS) Vol.4 No.9*

LEAD AUTHOR

Aug. 2013

- Modeled and evaluated energy efficiency of peer-assisted edge CDN architectures versus traditional centralized distribution.

Assessing The Dynamics of BitTorrent Swarm Topologies Using The Peer Exchange Protocol

*IEICE Transactions on
Communications (Special Issue on
Information Network Science,
Vol.E95-B, No.05)*

LEAD AUTHOR

2012

- Empirically inferred dynamic overlay topologies and peer discovery mechanisms using the Peer Exchange (PEX) protocol.

A Temporal View of The Topology of Dynamic BitTorrent Swarms

*IEEE INFOCOM Workshop on
Network Science for
Communication Networks (DOI:
10.1109/INFCOMW.2011.5928939)*

LEAD AUTHOR

2011

- Analyzed temporal graph evolution and topological robustness of large-scale distributed swarms under dynamic churn.

Education

Keio University

Tokyo, Japan

PH.D. IN MEDIA AND GOVERNANCE (INTERNET ARCHITECTURE & DISTRIBUTED SYSTEMS)

Graduated Oct. 2014

- Dissertation: Dynamic P2P network topologies, swarm dynamics, and energy-efficient Content Delivery Networks (CDNs).
- Supervisors: Prof. Jun Murai (Internet Hall of Fame / WIDE Project), Prof. Rodney Van Meter (Quantum Networking Pioneer), Dr. Kenjiro Cho, Prof. Hiroshi Kusumoto.

Institut Teknologi Bandung (ITB)

Bandung, Indonesia

M.T. (MASTER OF TECHNOLOGY IN ELECTRICAL ENGINEERING)

Graduated Oct. 2004

- Designed and benchmarked one of Indonesia's earliest IPv6 campus network backbones and core routing deployments.

Institut Teknologi Bandung (ITB)

Bandung, Indonesia

S.Si. (BACHELOR OF SCIENCE IN PHYSICS)

Graduated Oct. 2000

- Researched data assimilation methods in 1D hydrodynamic fluid systems (numerical modeling, wave equations, fluid mechanics).